Benchmarking Forecast Rankings for Deployment-Aware Model Selection

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Abstract

Forecasting benchmarks typically rank models by forecast-side metrics. In deployment, however, forecasts are often translated into actions through fixed operational interfaces, and the model preferred by forecast-side evaluation need not be the model that performs best after execution. We study this as a forecasting benchmark question: under a fixed frictional interface, does forecast-side ranking remain reliable for deployed model selection? We focus on Q2, the fixed-interface selection problem, and use Q1 only as mechanism support for why divergence can arise. Across a synthetic zero-friction anchor, a forecasting-native event-micro benchmark, a complementary Traffic-Hourly alert family, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal as friction increases. These results do not invalidate forecast-side evaluation for prediction; rather, they suggest that forecasting benchmarks aimed at deployment may need to report deployed-selection robustness alongside forecast-side metrics.

1. Introduction

Forecasting benchmarks typically compare models using forecast-side metrics. This is appropriate when the goal is to measure predictive quality. But many forecasting systems are not used directly: they are deployed through fixed interfaces that translate forecasts into actions under operational frictions such as switching costs, inertia, or execution constraints. In such settings, the model selected by forecast-side evaluation need not be the model that performs best after execution.

This paper studies that mismatch as a forecasting evaluation problem. Our main question is not how to learn a better policy, but whether forecast-side model ranking remains reliable for deployment-aware model selection under one fixed forecast-to-decision interface. We call this Q2. We separate Q2 from a mechanism question, Q1, which asks whether the same proposed action path can produce different realized outcomes under different interfaces. This separation keeps forecast-side metrics in their role as prediction measures while isolating when the object of model selection changes under deployment friction.

Forecasting benchmark work often evaluates forecast quality, calibration, and live predictive performance. Our question is complementary: whether forecast-side ranking remains trustworthy for deployed model selection under a fixed frictional interface. Unlike decision-aware forecasting or control work, we do not propose new training objectives or policies; we study the evaluation object used for model selection after forecasts are mediated by actions and friction.

This paper makes three narrow contributions.

- First, we introduce deployed-selection robustness as a forecasting benchmark question: whether forecast-side model ranking remains reliable after forecasts are executed through a fixed interface.
- Second, we document a Q2 failure mode: the forecast-side winner becomes recurrently deployed-suboptimal as friction increases under the studied fixed interfaces.
- Third, we provide a Q2-first evidence hierarchy spanning synthetic, forecasting-native event and traffic evidence, and operational corroboration; Q1 is retained only as mechanism support.

2. Two Evaluation Questions

We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. Forecast-side quality measures predictive performance, target-side decision quality scores the proposed action path before frictional execution, and realized-action quality scores the executed path after the fixed interface is applied. Q1 holds the proposed action path fixed and varies only the interface, so it is an exact-control mechanism check. Q2 holds the interface fixed and varies the forecaster, so it is the benchmark-selection question. Because Q2 asks whether the forecast-side winner remains the deployed winner, Q2 is the paper's main empirical claim.

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Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q2	interface	forecaster	aligned at zero	ranking flips	zero-fric. anchor
Event micro	Q2	interface	forecaster	mixed at zero	selection failure recurs	main fcst.-native Q2 evid.
Traffic Top-k Alert	Q2	interface	forecaster	aligned at zero	switching-cost divergence recurs	comp. second fcst.-native family
Traffic Relative-Rank	Q2	interface	forecaster	aligned at zero	qualitative separation recurs	appx.-only fcst.-native support
Inventory	Q2	interface	forecaster	mixed at zero	stronger recurrence	main oper. corrob.
Load-following	Q2	interface	forecaster	mixed at zero	directional recurrence	appx.-only sec. corrob.
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	mech. support
Inventory	Q1	proposal	interface	coincide at zero	threshold story	mech. support

Table 1. Q2-first evidence hierarchy. Synthetic provides the zero-friction anchor, event-micro and Traffic-Hourly provide forecasting-native Q2 evidence, inventory provides operational corroboration, and Q1 remains mechanism support.

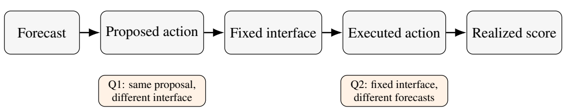

Figure 1. Forecast-to-decision evaluation setup. Forecasts are converted into proposed actions and then executed through a fixed frictional interface. Q2, the paper's main question, asks whether forecast-side model ranking matches deployed model ranking under one fixed interface. Q1 is used only as mechanism support.

3. Evidence Design

Our evidence hierarchy is Q2-first. Synthetic provides an exact zero-friction sanity anchor. Event-micro provides the main forecasting-native Q2 evidence: forecasters emit event probabilities, a fixed threshold maps probabilities to actions, and switching friction penalizes action changes. Traffic-Hourly Top-k provides a second forecasting-native task family under the same fixed-interface selection question. Inventory provides operational corroboration, while load-following and broader inventory sweeps are retained as appendix-only transparency checks. Across domains, the central quantity is whether the forecast-side winner remains the deployed winner under the fixed interface.

In inventory Q2, the main text retains only the 0.50 and 1.00 rows from the fixed five-family comparison; the mixed 0.00/0.25 regime and broader family sweep remain appendix-only transparency checks.

4. Results

4.1. Q2: Fixed interface, benchmark failure and selection consequence

Q2 asks the paper's headline benchmark question: under one fixed forecast-to-decision interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic provides the exact zero-friction sanity anchor; Figure 2 focuses on the fixed-interface winner-inversion evidence in the forecasting-native and corroborative domains.

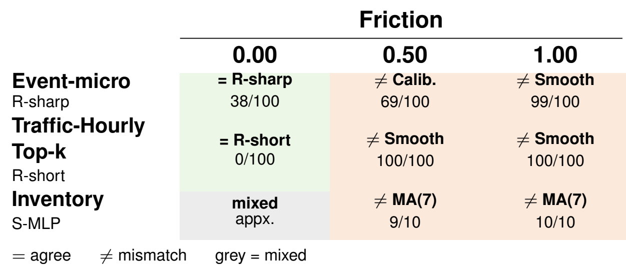

Figure 2. Forecast-side winners need not remain deployed winners. Across fixed-interface forecasting tasks, forecast-side ranking and deployed selection agree at or near zero friction but separate as switching friction increases. The main benchmark consequence is deployed-suboptimal selection: the model selected by forecast-side metrics is recurrently not the deployed-best model. Inventory's zero-friction cell is marked as mixed and is not used as headline corroboration.

In event-micro, the forecast-side winner remains Reactive sharp across frictions, but the deployed winner shifts to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The benchmark consequence is recurrent: the forecast-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The same qualitative inversion persists under an alternate threshold, under a hysteresis-threshold interface, and when forecast-side ranking is recomputed with log loss.

Friction	Forecast winner	Deployed winner	Agree.	Subopt. seeds
0.00	Reactive sharp	Reactive sharp	0.62	38/100
0.50	Reactive sharp	Calibrated baseline	0.31	**69/100**
1.00	Reactive sharp	Lagged smoother	0.01	**99/100**

Table 2. Event-micro Q2 evidence. Under a fixed thresholding interface, the forecast-side winner remains Reactive sharp, but the deployed winner shifts as switching friction increases. The forecast-selected model becomes deployed-suboptimal in 69/100 and 99/100 seeds at moderate and high friction.

A complementary Traffic-Hourly fixed-budget alert family shows the same benchmark consequence under a different forecasting-native task structure. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality reaches 100/100 seeds at both friction 0.5 and friction 1.0. A relative-rank Traffic variant shows the same qualitative pattern in the appendix.

Inventory provides operational corroboration rather than the main forecasting-native evidence. At friction 0.50 and 1.00, the forecast-selected Small MLP is deployed-suboptimal in 9/10 and 10/10 seeds, respectively, under the fixed replenishment interface.

Domain	Friction	Forecast winner	Deployed winner	Subopt. seeds
Traffic-Hourly Top-k	0.00	Reactive short	Reactive short	0/100
Traffic-Hourly Top-k	0.50	Reactive short	Lagged smoother	100/100
Traffic-Hourly Top-k	1.00	Reactive short	Lagged smoother	100/100
Inventory	0.50	Small MLP	Moving average (7)	9/10
Inventory	1.00	Small MLP	Moving average (7)	10/10

Table 3. Forecasting-native and operational corroboration. Traffic-Hourly provides a second forecasting-native task family, while inventory provides operational corroboration. Both show that forecast-side winners can become deployed-suboptimal under fixed interfaces at moderate-to-high friction.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

Appendix robustness. Detailed threshold, reranking, and second-interface checks for event-micro are retained in the appendix together with the Traffic relative-rank redesign, the broader inventory sweep, and the load-following transparency domain. Those appendix-only checks preserve the same narrow reading without changing the paper's main claim: under a fixed interface, a forecast-side winner can remain prediction-best while becoming recurrently deployed-suboptimal once switching friction is introduced.

Why forecast ordering need not be preserved. Proper forecast-side scores evaluate predictive quality, not the composed object obtained after forecasts are mapped to actions and executed through a frictional interface. If a forecaster's forecast-side advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. This is why Q1 is useful as mechanism support and why Q2 is the actual benchmark-selection question.

5. Discussion, Related Work, and Limitations

Related work. Our work is closest to forecasting evaluation and benchmark design, including live or leaderboard-style forecasting benchmarks such as ForecastBench and Prophet Arena and time-series evaluation frameworks such as GIFT-Eval (Karger et al., 2025; Yang et al., 2025; Aksu et al., 2024). It also sits alongside scoring-rule foundations and decision-maker-perspective forecast evaluation, which distinguish predictive quality from downstream use (Gneiting and Raftery, 2007; Murphy, 1993; Raeth and Ludwig, 2025). Unlike these works, we study whether forecast-side model ranking remains reliable for deployed model selection once forecasts pass through one fixed frictional interface. Our contribution is evaluative rather than algorithmic: we do not learn a policy or modify the training objective. We contrast with decision-focused learning and predict-then-optimize only to clarify scope; unlike these methods, we keep training fixed and ask what a forecasting benchmark should report for deployment-facing model selection (Donti et al., 2017; Elmachtoub and Grigas, 2022).

Discussion and limitations. Our claim is evaluative and deliberately narrow. We do not claim that forecast-side metrics are invalid for prediction, and we do not seek a single reporting rule for all forecasting applications. Event-micro provides the main forecasting-native evidence, Traffic-Hourly provides a second forecasting-native task family, and inventory provides operational corroboration, but this is still not a broad benchmark suite. Load-following and broader inventory sweeps remain appendix-only transparency checks, and the strongest main-text mismatches arise in the moderate-to-high friction settings studied here. Our results therefore support a reporting recommendation: deployment-facing forecasting benchmarks should test deployed-selection robustness under fixed interfaces alongside standard forecast-side metrics.

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A. Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A1. Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from $38/100$ seeds at zero friction to $99/100$ at friction 1.00.

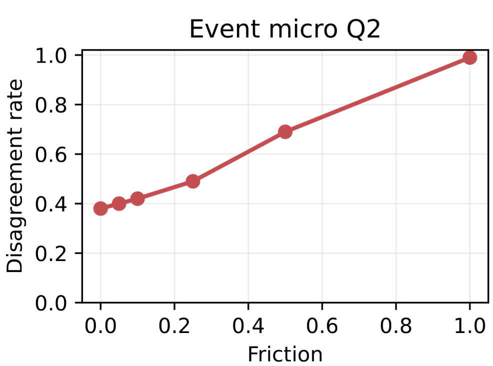

Figure A1. Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A2. Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A3. Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A4. Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

A.1. Interface-Aware Event-Micro Stats Addendum

Because the hysteresis-threshold check is retained paper-facing as a second-interface robustness result, this addendum records interface-specific recurrence and gap summaries without changing the main cross-domain recurrence package.

Interface	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Fixed threshold	0.50	69/100	0.69	[0.605, 1.000]	¡0.001
Fixed threshold	1.00	99/100	0.99	[0.953, 1.000]	¡0.001
Hysteresis threshold	0.50	55/100	0.55	[0.463, 1.000]	0.184
Hysteresis threshold	1.00	82/100	0.82	[0.745, 1.000]	¡0.001

Table A5. Interface-aware recurrence tests for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A6. Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

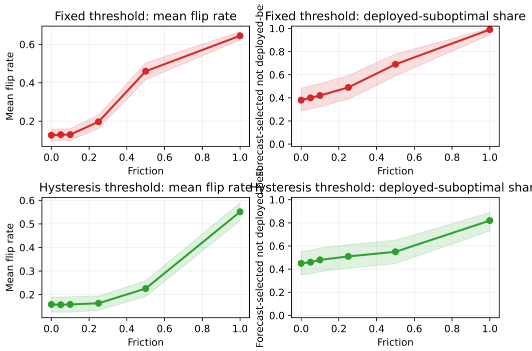

Figure A2. Interface-aware event-micro uncertainty addendum. Rows facet the canonical fixed threshold and the hysteresis threshold; columns show mean flip rate bands and deployed-suboptimal share bands.

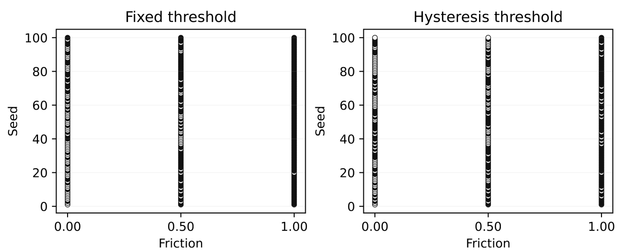

Figure A3. Interface-aware seed-level disagreement strips for the event-micro benchmark. Each panel shows the same seed set under one fixed interface.

B. Traffic-Hourly Forecasting-Native Redesign Families

B.1. Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A7. Forecast-side vs. deployed-side model selection in a complementary fixed-budget Traffic-Hourly alert family. The forecast-side winner is deployment-optimal at zero friction, but switching costs shift the deployed winner to a smoother family.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

B.2. Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A8. Relative-rank Traffic redesign retained as forecasting-native appendix support. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

The relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we retain this variant as appendix-only forecasting-native support. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

C. Inventory Operational Corroboration

This appendix records the full five-family inventory summary that underlies the compact corroboration rows in the main text. The mixed zero and low-friction rows are retained here for completeness, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table A9. Full inventory Q2 summary retained as appendix corroboration under a fixed replenishment interface.

D. Inventory Sweep Transparency

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid, fixed responsive interface, deployed metric, and tie policy unchanged. This appendix section is included as a broader transparency layer rather than as earlier or stronger main evidence than the five-family inventory comparison. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the main inventory Q2 pipeline.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	does not clear stricter gate

Table A10. Expanded same-interface inventory comparison summary retained for appendix transparency. These statuses do not overturn the narrower main-text claim from the five-family inventory comparison.

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	18	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	6	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	36	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	216	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	108	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	81	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A11. Standardized tuning protocol for the stronger-baseline comparison. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A13. Expanded-family inventory Q2 robustness summary for the broader comparison set.

In this broader comparison, agreement is mixed at 0.00 and 0.25, falls further by 0.50 and 1.00, deployed-suboptimal counts rise from $4/10$ to $9/10$, and moderate/high-friction mismatch strengthens. The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

E. Load-Following Secondary Corroboration

We retain aggregated electricity load-following only as appendix-only secondary corroboration under a fixed responsive interface. It is not part of the main forecasting-native evidence and is included only to show that the same Q2 direction can appear in a second operational setting. Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units.

A grouping-only balance-repair rerun substantially reduces group imbalance while preserving the Q2 drift structure. A final restricted cap/margin search, however, finds no jointly feasible configuration that also satisfies the remaining Q1 requirement, so the domain remains appendix-only secondary corroboration rather than additional main evidence.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A14. Load-following Q2 summary retained as appendix-only secondary corroboration under a fixed responsive interface.

This mixed zero row is not a tie-policy artifact: the zero-row winning sets are singletons in all seeds, yet the selected candidate still shows $3/10$ zero-friction mismatches with mean deployed gap 0.0002 and median gap 0.0. The grouping-only balance repair lowers the zero-row mean gap to about 0.0001 but does not remove the mismatch, so the most plausible reading is near-zero seed instability or grouping granularity rather than a clean zero-friction story.

From friction 0.25 onward, the forecast-side winner remains Linear AR while the deployed winner is Small MLP, agreement stays at 0.30, 0.30, and 0.20, and deployed-suboptimal counts rise to $7/10$, $7/10$, and $8/10$. The domain therefore contributes directionally consistent secondary corroboration only.

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=7, lasso, alpha=0.1	-2.418	-2.410
Gradient-boosted trees	lag=14, trees=100, depth=5, lr=0.1, leaf=10	-2.425	-2.279
Large MLP	w=256, d=3, drop=0.0, lr=0.001, wd=0.0001	-1.954	-2.012
GRU variant	L=28, h=128, layers=2, drop=0.0, lr=0.001	-2.602	-2.599

Table A12. Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only.

F. Q1 Mechanism Support

Q1 is retained only as mechanism support for why Q2 divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

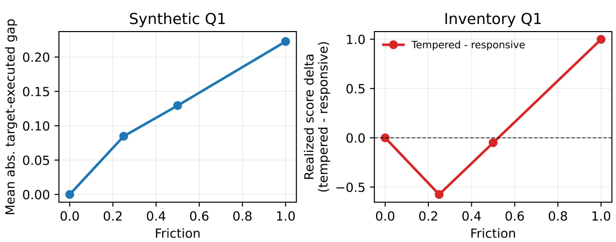

Figure A4. Appendix Q1 mechanism support. Synthetic yields exact zero-friction coincidence and a rising positive-friction proposal-versus-realization gap, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. The figure explains why Q2 divergence can arise but is not itself the paper's main evidence.

G. Additional Uncertainty and Seed Recurrence

This appendix collects cross-domain seed-level reliability checks for the main Q2 rows. These appendix-only additions do not overturn the narrower main-text claim.

Domain	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Event micro	0.50	69/100	0.69	[0.605, 1.000]	¡0.001
Event micro	1.00	99/100	0.99	[0.953, 1.000]	¡0.001
Inventory	0.50	9/10	0.90	[0.606, 1.000]	0.011
Inventory	1.00	10/10	1.00	[0.741, 1.000]	¡0.001
Load-following	0.25	7/10	0.70	[0.393, 1.000]	0.172
Load-following	0.50	7/10	0.70	[0.393, 1.000]	0.172
Load-following	1.00	8/10	0.80	[0.493, 1.000]	0.055

Table A15. One-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds. The strongest event-micro and inventory rows show clear majority recurrence, while load-following shows the same directional pattern with smaller seed counts.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Inventory	0.50	0.308	[0.186, 0.443]	0.227	[0.176, 0.484]
Inventory	1.00	1.187	[0.995, 1.390]	1.162	[0.969, 1.389]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A16. 95% bootstrap intervals for deployed-gap rows in the main Q2 evidence blocks. Event-micro and inventory provide the strongest positive-gap support, while load-following remains directional secondary corroboration.

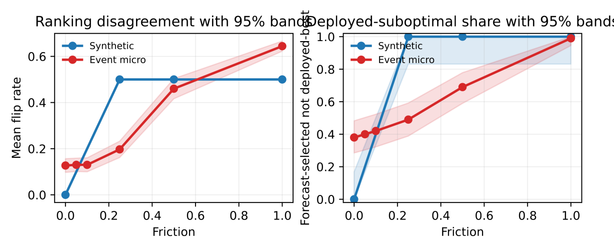

Figure A5. Appendix uncertainty bands for the synthetic and event-micro Q2 curves. Left: mean flip rate with 95% bands from seed-level standard errors. Right: deployed-suboptimal share with exact 95% binomial intervals.

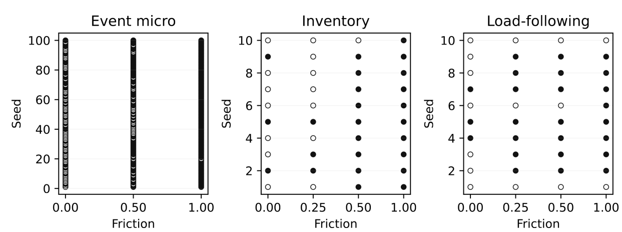

Figure A6. Seed-level disagreement strips for the main Q2 domains. Filled markers indicate seeds where the forecast-selected model is not deployed-best under the fixed interface.